Comparing a Score of 92

AP Statistics · Unit 2 · Topic 2.11 · TEACHER KEY

CED TETHER

- SKILLS: Using Probability and Simulation
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 3.C** — Describe probability distributions.
- ENDURING UNDERSTANDING: VAR-6 The normal distribution may be used to model variation.
- VAR-6.A Calculate the probability that a particular value lies in a given interval of a normal distribution. [Skill 3.A]
- VAR-6.A.1 A continuous random variable is a variable that can take on any value within a specified domain. Every interval within the domain has a probability associated with it. VAR-6.A.2 A continuous random variable with a normal distribution is commonly used to describe populations. The distribution of a normal random variable can be described by a normal, or "bell-shaped," curve. VAR-6.A.3 The area under a normal curve over a given interval represents the probability that a particular value lies in that interval.
- VAR-6.B Determine the interval associated with a given area in a normal distribution. [Skill 3.A]
- VAR-6.B.1 The boundaries of an interval associated with a given area in a normal distribution can be determined using z-scores or technology.
- VAR-6.C Determine the appropriateness of using the normal distribution to approximate probabilities for unknown distributions. [Skill 3.C]
- VAR-6.C.1 Normal distributions are symmetrical and "bell-shaped." As a result, normal distributions can be used to approximate distributions with similar characteristics.

Essential question: How can the same raw score have different relative standings in two normal distributions with the same mean?

DOK-3 skill: 4.B · **FRQ pattern:** `same-score-two-models-compare-with-z`

DOK rationale: Part (c) combines two percentile comparisons with the effect of spread to judge competing same-score claims and explain the relative standing supported by the models.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- How many School A standard deviations above 80 is a 92?
- Why does School B's wider curve place more area above the same score?
- Which area gives a percentile: left of 92 or right of 92?
- What exactly do you mean by 'better' when the raw score is identical?

[WARN] LOOK FOR

- Using the shared mean as evidence that all corresponding percentiles match.
- Reporting the upper-tail probability rather than the percentile below 92.
- Reversing the comparison and assigning the higher percentile to the wider distribution.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Supports the data coach using percentiles about 90.9 at A and 78.8 at B.
- **judgment-2** (required) — Explains that the same 12-point gap is different numbers of standard deviations because the spreads differ.
- **judgment-3** (required) — Concludes that 92 has higher relative standing at A and explains why the shared mean is insufficient.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

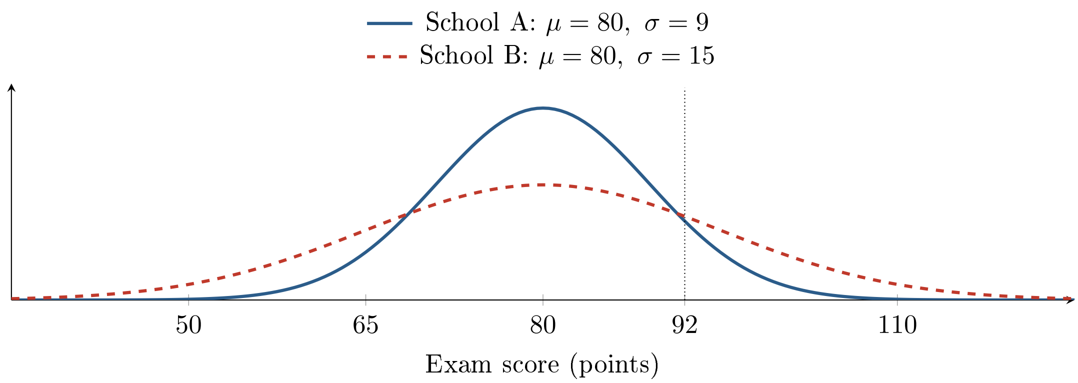
Common mistakes

- Using 80 as the standard deviation because it is shared by both models
- Computing $z = (80 - 92)/\sigma$ and reporting lower-tail areas without correcting direction
- Saying the wider distribution makes every above-mean score more exceptional
- Calling one raw score objectively higher even though the numerical score is 92 at both schools

[STAR] THE PROBLEM — annotated

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE — one sentence before you work the parts

Would a 92 stand out equally at both schools? Give one reason from the picture.
Key: Not graded. Equal means should make the counselor’s claim tempting; standardization should change the final judgment.

[BOOK] STANDARDIZE BEFORE COMPARING (keep on the page)

For a normal random variable, $z = \dfrac{x-\mu}{\sigma}$ tells how many standard deviations x is from its mean. The percentile of x is the percent of values at or below x. Use a normal table or calculator to find the area to the left of the z-score; multiply that area by 100. A larger percentile means a higher standing relative to that school’s scores. Printed normal table: $P(Z \le 0.8000) = 0.788145$ and $P(Z \le 1.3333) = 0.908789$. These left areas supply the lookup needed above.

DOK 1 **(a)** Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

Key: School B is more variable because $\sigma_B = 15$ points is larger than $\sigma_A = 9$ points. Its normal curve is wider and lower; School A’s is narrower and taller around the common mean of 80.

DOK 2 **(b)** For a score of 92, calculate the z-score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

Key: School A: $z = (92-80)/9 = 1.333\ldots$, so $P(X_A \le 92) = \Phi(1.333\ldots) \approx 0.9088$ and 92 is at about the 90.9th percentile. School B: $z = (92-80)/15 = 0.80$, so $P(X_B \le 92) = \Phi(0.80) \approx 0.7881$ and 92 is at about the 78.8th percentile.

DOK 3 **(c)** In 3–4 sentences, decide whose claim is better supported. Use both percentiles from (b) and explain how the different spreads change what a score of 92 means at each school.

Key: The data coach’s claim is better supported. A 92 is about the 90.9th percentile at A and the 78.8th percentile at B. The same 12-point gap is 1.33 standard deviations at A but only 0.80 at B because the spreads differ. Thus a 92 stands higher relative to scores at School A; the shared mean alone does not settle the comparison.